

# Manish Raghavan

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<https://mraghavan.github.io/>

## EMPLOYMENT

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### Drew Houston (2005) Career Development Professor

August 2022–Present

*Massachusetts Institute of Technology, Cambridge, MA*

*Sloan School of Management and Department of Electrical Engineering and Computer Science*

### Postdoctoral Fellow

August 2021–August 2022

*Harvard Center for Research on Computation and Society, Cambridge, MA*

### Visiting Researcher

September 2019–August 2021

*Facebook, New York, NY*

## EDUCATION

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### Ph.D., Computer Science

*Cornell University*

Thesis: *The Societal Impacts of Algorithmic Decision-Making*. Advised by Jon Kleinberg.

### M.S., Computer Science

*Cornell University*

### B.S., Electrical Engineering and Computer Science

*University of California, Berkeley*

## PUBLICATIONS

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### Conferences

- N. Dalvi, M. Olteanu, M. Raghavan, and P. Bohannon. *Deduplicating a Places Database*. In *Proceedings of the 23rd International World Wide Web Conference*. April 2014
- J. Kleinberg, S. Oren, and M. Raghavan. *Planning Problems for Sophisticated Agents with Present Bias*. In *Proceedings of the 17th ACM Conference on Economics and Computation*. July 2016
- J. Kleinberg, S. Mullainathan, and M. Raghavan. *Inherent Trade-offs in the Fair Determination of Risk Scores*. In *The 8th Innovations in Theoretical Computer Science Conference*. January 2017
- J. Kleinberg, S. Oren, and M. Raghavan. *Planning with Multiple Biases*. In *Proceedings of the 18th ACM Conference on Economics and Computation*. June 2017
- G. Pleiss, M. Raghavan, F. Wu, J. Kleinberg, K. Weinberger. *On Fairness and Calibration*. In *Proceedings of the 31st Annual Conference on Neural Information Processing Systems*. December 2017
- J. Kleinberg, M. Raghavan. *Selection Problems in the Presence of Implicit Bias*. In *The 9th Innovations in Theoretical Computer Science Conference*. January 2018
- M. Raghavan, A. Anderson, J. Kleinberg. *Mapping the Invocation Structure of Online Political Interaction*. In *Proceedings of the 27th International World Wide Web Conference*. April 2018
- M. Raghavan, A. Slivkins, J. W. Vaughan, Z. S. Wu. *The Externalities of Exploration and How Data Diversity Helps Exploitation*. In *Conference on Learning Theory*. July 2018
- M. Raghavan, M. Purohit, S. Gollapudi. *Hiring Under Uncertainty*. In *International Conference on Machine Learning*. June 2019
- R. Abebe, S. Barocas, J. Kleinberg, K. Levy, M. Raghavan, D. G. Robinson. *Roles for Computing in Social Change*. In *Proceedings of the Third ACM Conference on Fairness, Accountability, and Transparency*. January 2020

- S. Barocas, A. D. Selbst, M. Raghavan. *The Hidden Assumptions Behind Counterfactual Explanations and Principal Reasons*. In *Proceedings of the Third ACM Conference on Fairness, Accountability, and Transparency*. January 2020
- M. Raghavan, S. Barocas, J. Kleinberg, K. Levy. *Mitigating Bias in Algorithmic Hiring: Evaluating Claims and Practices*. In *Proceedings of the Third ACM Conference on Fairness, Accountability, and Transparency*. January 2020
- J. Finocchiaro, R. Maio, F. Monachou, G. K. Patro, M. Raghavan, A.-A. Stoica, and S. Tsirtis. *Bridging Machine Learning and Mechanism Design towards Algorithmic Fairness*. In *Proceedings of the Fourth ACM Conference on Fairness, Accountability, and Transparency*. January 2021
- J. Kleinberg, S. Oren, M. Raghavan, and N. Sklar. *Stochastic Model for Sunk Cost Bias*. In *Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence*. July 2021
- E. Black, M. Raghavan, and S. Barocas. *Model Multiplicity: Opportunities, Concerns, and Solutions*. In *Proceedings of the Fifth ACM Conference on Fairness, Accountability, and Transparency*. June 2022
- C. Hays, Z. Schutzman, M. Raghavan, E. Walk, P. Zimmer. *Simplistic Collection and Labeling Practices Limit the Utility of Benchmark Datasets for Twitter Bot Detection*. In *The Web Conference*. April 2023
- R. Alur, L. Laine, D. Li, M. Raghavan, D. Shah, D. Shung. *Auditing for Human Expertise*. In *Proceedings of the 37th Annual Conference on Neural Information Processing Systems*. December 2023
- C. Dwork, C. Hays, J. Kleinberg, M. Raghavan. *Content Moderation and the Formation of Online Communities: A Theoretical Framework*. In *The Web Conference*. May 2024
- K. Peng, M. Raghavan, E. Pierson, J. Kleinberg, N. Garg. *Reconciling the Accuracy-Diversity Trade-Off in Recommendations*. In *The Web Conference*. May 2024
- C. Dwork, C. Hays, J. Kleinberg, M. Raghavan. *Equilibria, Efficiency, and Inequality in Network Formation for Hiring and Opportunity*. In *Proceedings of the 25th ACM Conference on Economics and Computation*. July 2024
- R. Alur, M. Raghavan, D. Shah. *Human Expertise in Algorithmic Prediction*. In *Proceedings of the 38th Annual Conference on Neural Information Processing Systems*. December 2024
- B. Laufer, M. Raghavan, S. Barocas. *What Constitutes a Less Discriminatory Algorithm?*. In *ACM Symposium on Computer Science and Law*. March 2025
- Maria Ballesteros, Cynthia Dwork, Gary King, Conlan Olson, Manish Raghavan. *Evaluating the Impacts of Swapping on the US Decennial Census*. In *ACM Symposium on Computer Science and Law*. March 2025
- C. Hays, M. Raghavan. *Double Machine Learning for Causal Inference under Shared-State Interference*. In *International Conference on Machine Learning*. July 2025
- N. Jo, K. Creel, A. Wilson, M. Raghavan. *Homogeneous Algorithms Can Reduce Competition in Personalized Pricing*. In *Neural Information Processing Systems*. December 2025
- D. Shanmugam, S. Sadhuka, M. Raghavan, J. Gutttag, B. Berger, E. Pierson. *Evaluating multiple models using labeled and unlabeled data*. In *Neural Information Processing Systems*. December 2025
- C. Hays, B. Laufer, S. Barocas, M. Raghavan. *Statistical Guarantees in the Search for Less Discriminatory Algorithms*. In *International Conference on Learning Representations*. April 2026
- C. Park, K. Donahue, M. Raghavan. *When to Ask a Question: Understanding Communication Strategies in Generative AI Tools*. In *Foundations of Responsible Computing*. June 2026
- C. Hays, R. Li, B. Flanigan, M. Raghavan. *Strategic Candidacy in Generative AI Arenas*. In *International Conference on Machine Learning*. July 2026
- N. Jo, N. Garg, M. Raghavan. *The Subjectivity of Monoculture*. In *Neural Information Processing Systems*. December 2026
- K. Ren, M. Raghavan, N. Garg. *Hitting a Moving Target: Test-Time Adaptation for AI Text Detection under Continual Distribution Shift*. In *Neural Information Processing Systems*. December 2026
- K. Donahue, M. Raghavan. *Impacts of Aggregation on Model Diversity and Consumer Utility*. In *Neural Information Processing Systems*. December 2026

## Journals

- J. Kleinberg, M. Raghavan. *How Do Classifiers Induce Agents To Invest Effort Strategically?*. In *ACM Transactions on Economics and Computation*. October 2020

Also appeared as:

*How Do Classifiers Induce Agents To Invest Effort Strategically?*. In *Proceedings of the 20th ACM Conference on Economics and Computation*. June 2019

*Designing Evaluation Rules that are Robust to Strategic Behavior*. In *Proceedings of the 34th AAAI Conference on Artificial Intelligence, Sister Conference Track*. April 2020

*Algorithmic Classification and Strategic Effort*. In *SIGecom Exchanges*. November 2020

- J. Kleinberg and M. Raghavan. *Algorithmic Monoculture and Social Welfare*. In *Proceedings of the National Academy of Sciences* 118(22). June 2021
- M. Raghavan, A. Slivkins, J. W. Vaughan, Z. S. Wu. *Greedy Algorithm Almost Dominates in Smoothed Contextual Bandits*. In *SIAM Journal on Computing*. April 2023
- J. Kleinberg, S. Mullainathan, and M. Raghavan. *The Challenge of Understanding What Users Want: Inconsistent Preferences and Engagement Optimization*. In *Management Science*. November 2023

Also appeared as:

*The Challenge of Understanding What Users Want: Inconsistent Preferences and Engagement Optimization*. In *Proceedings of the 23rd ACM Conference on Economics and Computation*. June 2022

- C. K. Morewedge, S. Mullainathan, H. F. Naushan, C. R. Sunstein, J. Kleinberg, M. Raghavan, and J. O. Ludwig. *Human Bias in Algorithm Design*. In *Nature Human Behavior*. November 2023
- J. Kleinberg, J. Ludwig, S. Mullainathan, and M. Raghavan. *The Inversion Problem: Why Algorithms Should Infer Mental State and Not Just Predict Behavior*. In *Perspectives on Psychological Science*. December 2023
- M. Raghavan and P. T. Kim. *Limitations of the “Four-Fifths Rule” and Statistical Parity Tests for Measuring Fairness*. In *Georgetown Law Technology Review*. January 2024
- M. Raghavan. *What Should We Do when Our Ideas of Fairness Conflict?*. In *Communications of the ACM*. January 2024
- C. Dwork, K. Greenewald, M. Raghavan. *Synthetic Census Data Generation via Multidimensional Multiset Sum*. In *Journal of Privacy and Confidentiality*. August 2025

Also appeared as:

*Synthetic Census Data Generation via Multidimensional Multiset Sum*. In *Foundations of Responsible Computing*. June 2024

- V. M. Suriyakumar, R. Alur, A. Sekhari, M. Raghavan, A. C. Wilson. *Unstable Unlearning: The Hidden Risk of Concept Resurgence in Diffusion Models*. In *Transactions on Machine Learning Research*. April 2026
- B. Hedden and M. Raghavan. *Algorithmic Monoculture and its Critics*. In *Philosophical Perspectives*. September 2026

## Other

- M. Olteanu, N. Dalvi, and M. Raghavan. *Identifying descriptive terms associated with a physical location from a location store*. In *U.S. Patent No. 9613054*. April 2017
- M. Raghavan, S. Barocas. *Challenges for mitigating bias in algorithmic hiring*. In *The Brookings Institution*. December 2019

## TEACHING

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**6.3950/6.3952: Artificial Intelligence, Decision-Making, and Society.** *MIT*

Fall 2022 (with Aleksander Mądry and Ashia Wilson); Fall 2023 (with Ashia Wilson)

**15.563: Artificial Intelligence for Business.** *MIT*

Spring 2023; Spring 2024; Spring 2026

**24/6.C40: Ethics of Computing.** *MIT*

Fall 2025 (with Brad Skow)

## AWARDS AND HONORS

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Cornell University Fellowship

2016–2017

NSF GRFP Fellowship

2017–2021

|                                                                          |           |
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| Microsoft Research PhD Fellowship                                        | 2018–2020 |
| SIGecom Doctoral Dissertation Award Honorable Mention                    | 2021      |
| ACM Doctoral Dissertation Award                                          | 2021      |
| Cornell University Department of Computer Science PhD Dissertation Award | 2022      |
| ACM EC Exemplary Applied Modeling Track Paper Award                      | 2022      |
| The Web Conference Best Paper Award                                      | 2023      |

## INVITED TALKS

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### Planning Problems for Sophisticated Agents with Present Bias

*Social Impact through Network Science.* Venice, Italy June 8–10, 2016  
*Young Researcher Workshop on Economics and Computation.* Tel Aviv, Israel January 1–5, 2017

### Algorithmic Fairness and Bias

*Deloitte Data Scientist Speaker Series.* Virtual September 7, 2018

### Mitigating Bias in Algorithmic Hiring: Evaluating Claims and Practices

*Fairness and Equity in Hiring Selection Algorithms.* Upturn, Washington D.C. October 4, 2019  
*ASSA Session on Auditing and Regulating AI Systems.* Virtual January 7, 2022  
*5th IDSC of IZA Workshop: Matching Workers and Jobs Online (Keynote).* Virtual September 16, 2022

### Algorithmic Fairness in Practice

*Walmart Community of Practice.* Virtual August 26, 2020

### The Societal Impacts of Algorithmic Decision-Making

*Netflix Research Seminar.* May 21, 2021  
*Sai University Frontiers Seminar.* Virtual November 23, 2021

### The Challenge of Understanding What Users Want: Inconsistent Preferences and Engagement Optimization

*NYU Data Science Seminar.* Virtual April 6, 2022  
*Harvard EconCS Seminar.* Virtual April 8, 2022  
*Facebook Core Data Science Seminar.* Virtual May 6, 2022  
*MIT Initiative on the Digital Economy seminar.* Virtual October 4, 2022  
*INFORMS Annual Meeting.* Indianapolis, IN October 18, 2022  
*Opportunity Insights.* Cambridge, MA October 26, 2022  
*JHU Behavioral Science Forum on Technology and Social Change.* Baltimore, MD October 28, 2022  
*UIUC Responsible Data Science and AI seminar.* Virtual November 4, 2022  
*INFORMS Annual Meeting.* Phoenix, AZ October 16, 2023

### Measuring Fairness

*FinRegLab Advisory Board.* Virtual July 14, 2021

### Understanding Societal Impacts through Machine Learning and Mechanism Design: Automated Hiring as a Case Study

*Israel Algorithmic Game Theory Seminar.* Virtual October 19, 2021

### The Future of AI in Financial Services (Panel)

*Artificial Intelligence and the Economy.* U.S. Dept. of Commerce, Washington D.C. April 27, 2022

### Reflections on Incentivizing Behavior through Evaluation

*Algorithmic Contract Design: Present and Future (EC '22 Workshop).* Boulder, CO July 15, 2022

### **Responsible AI and the Role of Measurement**

|                                                  |                    |
|--------------------------------------------------|--------------------|
| <i>Network Architecture for Geeks.</i> Virtual   | September 21, 2022 |
| <i>Sai University Frontiers Seminar.</i> Virtual | November 17, 2022  |
| <i>Wells Fargo Data Ethics.</i> Virtual          | April 26, 2023     |
| <i>Fujitsu x MIT Knowledge Session.</i> Virtual  | May 16, 2023       |

### **Is AI the New HR? Protecting Civil Rights at Work (Panel)**

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| <i>South by Southwest.</i> Austin, TX | March 11, 2023 |
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### **Generative AI and Employment**

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| <i>MIT Generative AI week.</i> Cambridge, MA | November 30, 2023 |
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### **Bias & discrimination in algorithmic decision-making**

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| <i>40th Annual NABE Economic Policy Conference.</i> Washington DC | January 16, 2024 |
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### **Human Expertise in Algorithmic Prediction**

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| <i>Bridging Prediction and Intervention Problems in Social Systems.</i> Banff, Canada                                         | June 2, 2024       |
| <i>Humans, Algorithmic Decision-Making and Society: Modeling Interactions and Impact (ICML '24 Workshop).</i> Vienna, Austria | July 27, 2024      |
| <i>TTIC Workshop on Data-Driven Decision Processes.</i> Chicago, IL                                                           | August 28, 2024    |
| <i>NSF-SDM Workshop on Human-AI Teaming.</i> Pittsburgh, PA                                                                   | September 26, 2024 |

### **Synthetic Data and Swapping for the US Decennial Census**

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| <i>CS+Law Virtual Workshop Series.</i> Virtual | November 15, 2024 |
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### **The Role of Information in Human-AI Collaboration**

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| <i>Tuck Operations and Management Science Seminar.</i> Hanover, NH                            | December 5, 2024 |
| <i>Simons Collaboration on Theory of Algorithmic Fairness.</i> New York, NY                   | February 6, 2025 |
| <i>UC Berkeley CLIMB Seminar.</i> Berkeley, CA                                                | March 20, 2025   |
| <i>UC Santa Barbara Technology Management distinguished speaker series.</i> Santa Barbara, CA | May 16, 2025     |

### **Competition and Diversity in Generative AI**

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| <i>SIGecom Winter Meeting.</i> Virtual                                                          | March 6, 2025      |
| <i>ICML Workshop on Reliable and Responsible Foundation Models (invited).</i> Vancouver, Canada | July 19, 2025      |
| <i>MIT Sloan Operations Management Seminar.</i> Cambridge, MA                                   | September 22, 2025 |
| <i>New Directions in Social Algorithms.</i> New Haven, CT                                       | October 17, 2025   |
| <i>MSR New England Econ+Comp/CSS Seminar.</i> Cambridge, MA                                     | December 9, 2025   |

## **SERVICE AND ORGANIZATION**

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| <b>International Advisory Board, Sai University (Chennai, India)</b> | 2021–Present |
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### **Program Committees**

- ACM Conference on Fairness, Accountability, and Transparency: 2020, 2021, 2022 (Area Chair), 2023 (Area Chair)
- ACM Conference on Economics and Computation: 2019, 2020, 2021, 2022
- ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization: 2021, 2022, 2023 (Area Chair)
- Conference on Neural Information Processing Systems: 2021
- Conference on Web and Internet Economics: 2021
- AAAI Conference on Artificial Intelligence: 2020, 2021

- International Conference on Learning Representations: 2021
- The Web Conference: 2024 (Senior Area Chair)

### **Workshops and Tutorials**

- Fairness and Discrimination through the Dual Lenses of Mechanism Design and Machine Learning (EC '21)
- Fair and Socially Responsible ML for Recommendations: Challenges and Perspectives (NeurIPS '22)

### **Journal Reviewing**

Journal of Machine Learning Research, Proceedings of the National Academy of Sciences, Artificial Intelligence, Management Science, Patterns, and others

### **Coaching**

*Harvard Men's Soccer Club*

2021–Present